Module 6-1 to 6-5 #1

Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

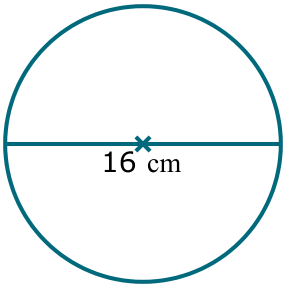
Number of Questions: **39**

Instructor Name: **Bodanske, Daniel**

Question 1 of 39

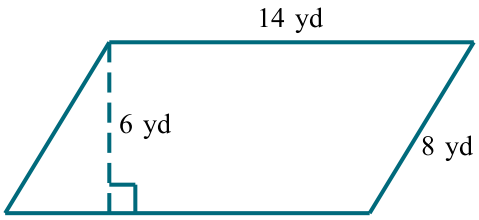
Find the area of a circle with diameter 16 cm.

Use the value 3.14 for π, and do not round your answer.
Be sure to include the correct unit in your answer.



Question 2 of 39

Find the area of this parallelogram. Be sure to include the correct unit in your answer.



_____

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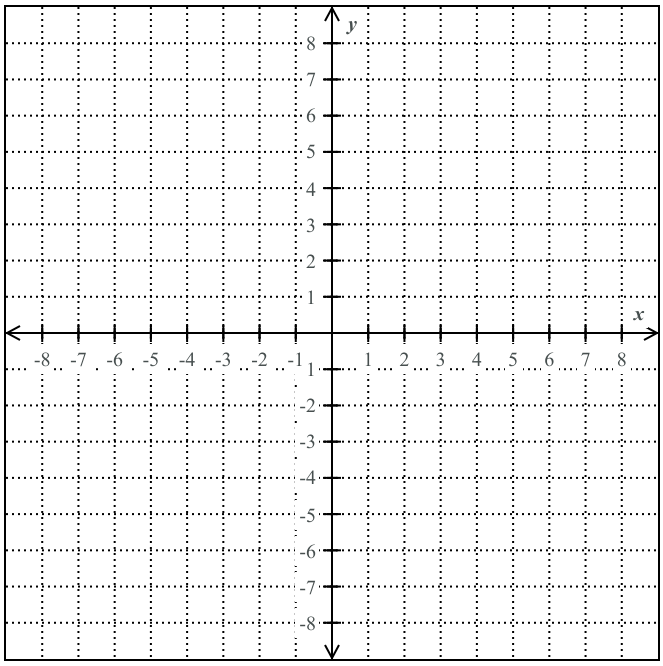
Question 3 of 39

Find the area and perimeter of the rectangle with vertices $(-5,\ -3)$, $(-2,\ -3)$, $(-2,\ 1)$, and $(-5,\ 1)$.

Note that you can draw in the Scratch Area below, but it is not part of the answer.



Area: _____ square units

Perimeter: _____ units

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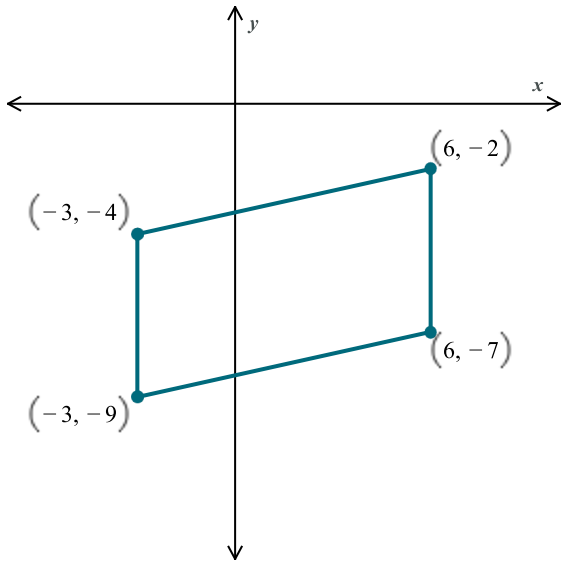
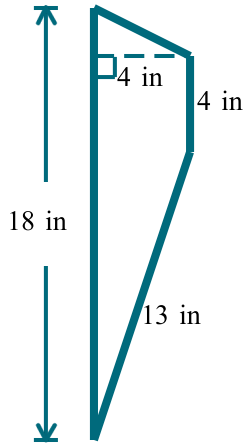
Question 4 of 39

What is the area of the parallelogram?



Question 5 of 39

Find the area of this trapezoid. Be sure to include the correct unit in your answer.



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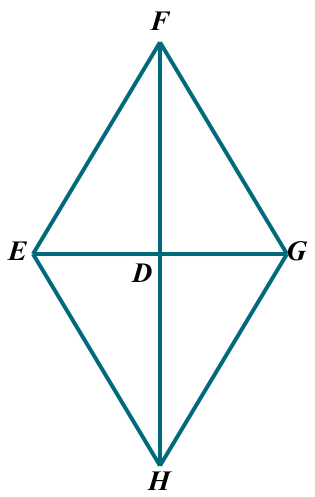
Question 6 of 39

In rhombus $EFGH$ below, $EG = 18$ mm and $FH = 30$ mm.
Find the area of the rhombus.

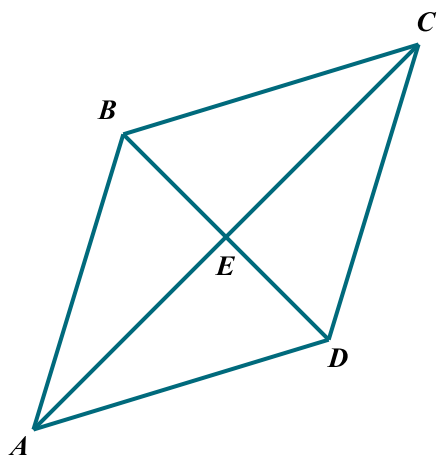
Be sure to include the correct unit in your answer.



Question 7 of 39

In rhombus $ABCD$ below, $AB = 17$ mm and $AE = 15$ mm.
Find the area of the rhombus.

Be sure to include the correct unit in your answer.



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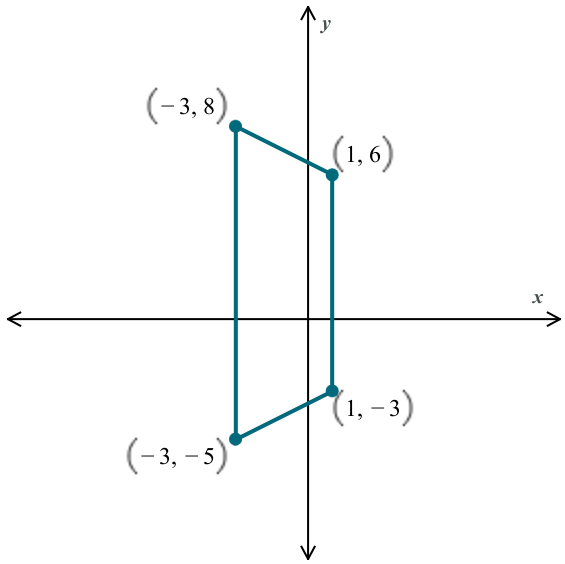
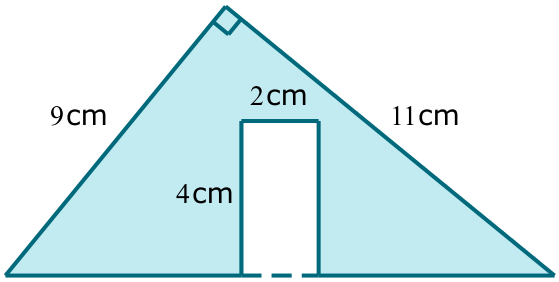
Question 8 of 39

What is the area of the trapezoid?



Question 9 of 39

A rectangle is removed from a right triangle to create the shaded region shown below.



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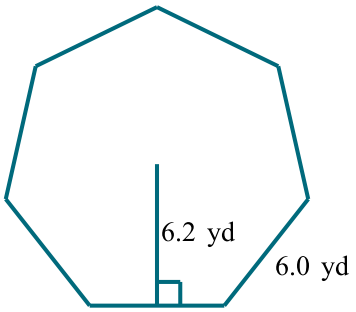
Question 10 of 39

A regular heptagon has a side of approximately $6.0\ \text{yd}$ and an apothem of approximately $6.2\ \text{yd}$. Find the area of the heptagon.



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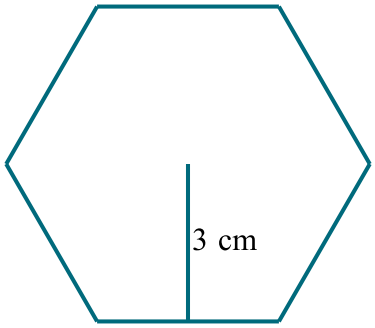
Find the area of a regular hexagon with apothem length $3\ \text{cm}$.
If necessary, write your answer in simplified radical form.



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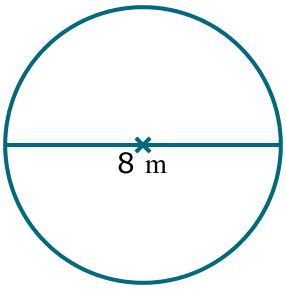
Question 12 of 39

Find the area of a circle with diameter $8\ \mathrm{m}$.

Use the value 3.14 for π, and do not round your answer.
Be sure to include the correct unit in your answer.



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Question 13 of 39

The circle below is divided into equal slices.
The figure to the right of the circle shows the slices rearranged.

As the number of slices increases, the rearranged figure more closely resembles a parallelogram.
For this problem, we will assume that the final rearranged figure is a parallelogram.
(We will refer to this parallelogram with quotes in the statements below.)

Suppose the circle has radius $r = 3$ cm.
Answer the following.
When applicable, use the symbol π for your answers. (Do not give decimal answers.)

Part 1:
The height of the "parallelogram" is {twice, half, equal to} the radius of the circle.
So, the height of the "parallelogram" is $h =$ _____ cm.

Part 2:
The length of the base of the "parallelogram" is {twice, half, equal to} the circumference of the circle.

The circumference of the circle is _____ cm.
So, the length of the base of the "parallelogram" is $b =$ _____ cm.

Part 3:
Use the parts above to find the area of the "parallelogram".

Area of "parallelogram" = _____ cm^2

Part 4:
The circle was rearranged to make the "parallelogram".
So the area of the circle is equal to the area of the "parallelogram".
Using this fact, choose the equation that gives the area of the circle in terms of r.

- ○ Area $= 9\pi r$
- ○ Area $= \pi r^2$
- ○ Area $= 6\pi r^2$
- ○ Area $= \pi r$
- ○ Area $= 3\pi r^2$

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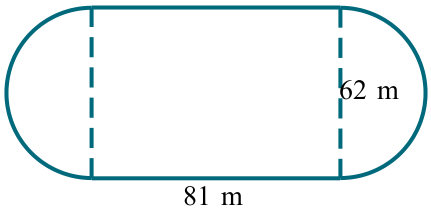
A training field is formed by joining a rectangle and two semicircles, as shown below. The rectangle is $81\ \mathrm{m}$ long and $62\ \mathrm{m}$ wide.

Find the area of the training field. Use the value 3.14 for π, and do not round your answer. Be sure to include the correct unit in your answer.



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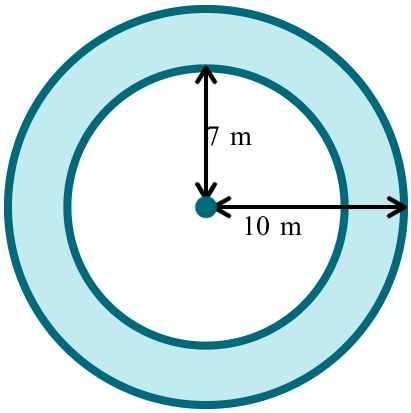
A ring-shaped region is shown below.
Its inner radius is $7\ \mathrm{m}$, and its outer radius is $10\ \mathrm{m}$.



Find the area of the shaded region.
Use 3.14 for π. Do not round your answer.

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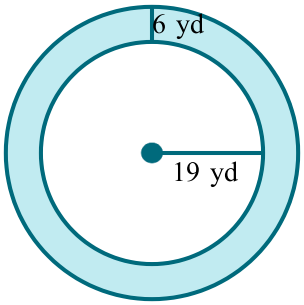
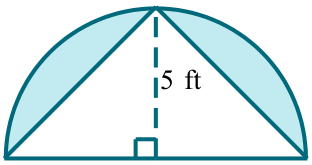
A flower garden is shaped like a circle. Its radius is $19\ \text{yd}$. A ring-shaped path goes around the garden. The width of the path is $6\ \text{yd}$.

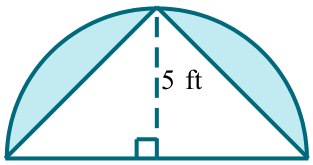


The gardener is going to cover the path with sand. If one bag of sand can cover $6\ \text{yd}^2$, how many bags of sand does the gardener need? Note that sand comes only by the bag, so the number of bags must be a whole number. (Use the value 3.14 for π.)

Question 17 of 39

A triangle is placed in a semicircle with a radius of $5\ \text{ft}$, as shown below. Find the area of the shaded region.
Use the value 3.14 for π, and do not round your answer. Be sure to include the correct unit in your answer.





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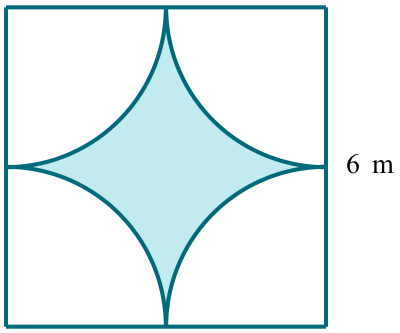
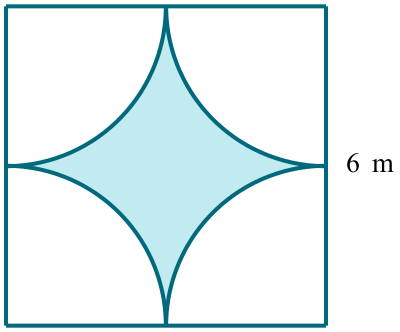
Four quarter circles are placed in a square with a side length of $6\ \mathrm{m}$, as shown below.

Find the area of the shaded region. Use the value 3.14 for π, and do not round your answer. Be sure to include the correct unit in your answer.





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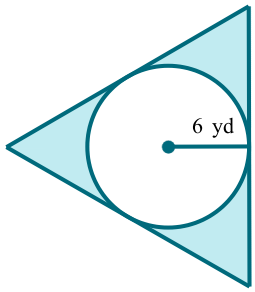
An equilateral triangle is circumscribed about a circle with a radius of 6 yd. Find the area of the shaded region shown.

Give the exact answer. (Do not approximate π or any square root.)



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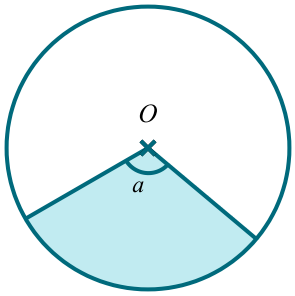
The circle has center O. Its radius is 5 yd, and the central angle a measures $110°$. What is the area of the shaded region?

Give the exact answer in terms of π, and be sure to include the correct unit in your answer.



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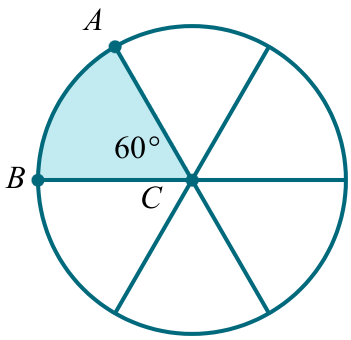
The circle below with center C is divided into 6 equal slices.

Central angle $\angle ACB$ intercepts $\overset{\frown}{AB}$, forming the shaded sector.



Complete the statements below.
Give your answers as exact values, not decimal approximations.

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(a) Find the ratio of the area of the sector to the area of the circle.

$$\frac{\text{Area of sector}}{\text{Area of circle}} = _____$$

(b) From the figure we see that $m\angle ACB = 60°$.

The measure of $\angle ACB$ is [twice / half / equal to] the measure of $\overset{\frown}{AB}$.

So, $m\overset{\frown}{AB} =$ _____.

This gives the following equation.

$$\frac{m\overset{\frown}{AB}}{360°} = _____$$

(c) Choose the equation that is true.

- ○ $\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{m\overset{\frown}{AB}}{360°}$
- ○ $\frac{\text{Area of sector}}{\text{Area of circle}} = 6 \cdot \frac{m\overset{\frown}{AB}}{360°}$
- ○ $\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{360°}{m\overset{\frown}{AB}}$
- ○ $\frac{\text{Area of sector}}{\text{Area of circle}} = m\overset{\frown}{AB}$

(d) Using πr^2 for the area of the circle, choose the equation that gives a formula for the area of the sector.

- ○ Area of sector $= \frac{360°}{m\overset{\frown}{AB}} \cdot \pi r^2$
- ○ Area of sector $= \frac{m\overset{\frown}{AB}}{360°} \cdot \pi r^2$
- ○ Area of sector $= \frac{m\overset{\frown}{AB}}{360°} \cdot 6\pi r^2$
- ○ Area of sector $= m\overset{\frown}{AB} \cdot \pi r^2$

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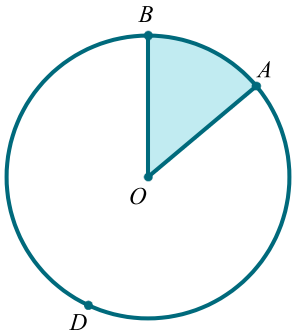
The circle below has center O, and its radius is 3 m. Given that $m\angle AOB = 50°$, find the length of the arc $\overset{\frown}{AB}$ and the area of the shaded region.

Give exact answers in terms of π, and be sure to include the correct units in your answer.



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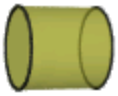
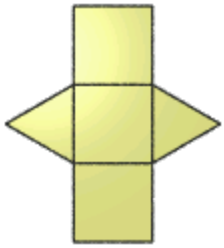
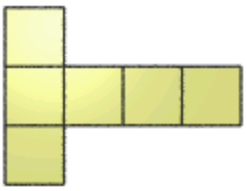
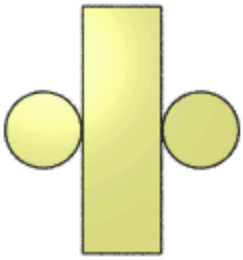
Here are three solids.

a.

b.

c. 

For each net below, select the solid above to which it folds.

1 ○ a ○ b ○ c ○ None	2  ○ a ○ b ○ c ○ None
3  ○ a ○ b ○ c ○ None	4  ○ a ○ b ○ c ○ None

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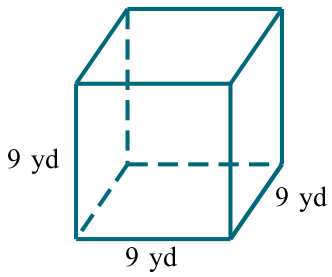
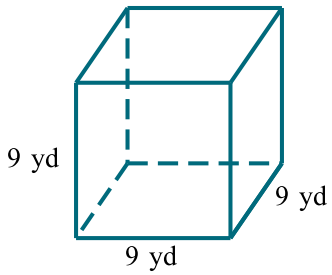
Find the surface area of this cube. Be sure to include the correct unit in your answer.





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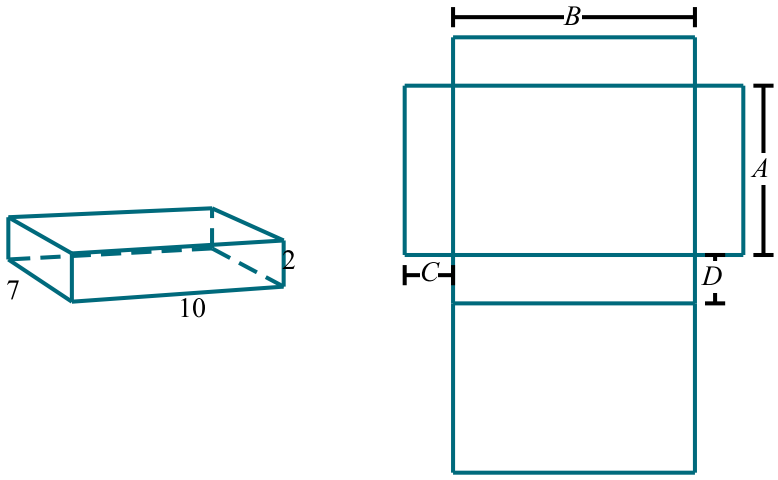
A rectangular prism and its net are shown below.
(All lengths are in feet.)



(a) Find the following side lengths for the net.

$A =$ _____ ft
$B =$ _____ ft
$C =$ _____ ft
$D =$ _____ ft

(b) Use the net to find the surface area of the prism.

_____ ft^2

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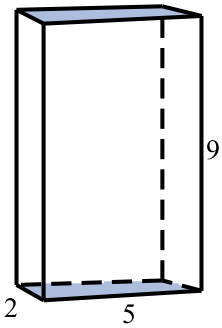
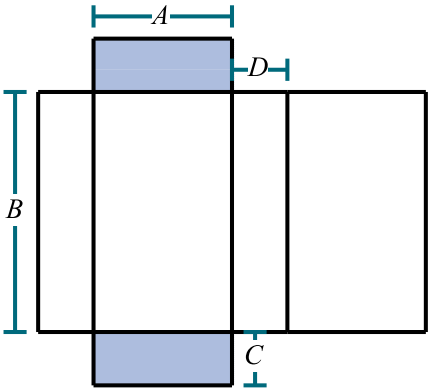
A rectangular prism and its net are shown below.
The top and bottom of the prism are shaded.
(All lengths are in meters.)





(a) Find the following side lengths for the net.

$A =$ m
$B =$ m
$C =$ m
$D =$ m

(b) Use the net to find the *lateral* surface area of the prism.
Neither the top nor bottom is included.

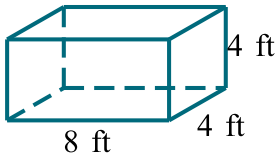
_____ m^2

(c) Use the net to find the *total* surface area of the prism.

_____ m^2

Question 27 of 39

Maria is going to build some large wooden storage boxes. The boxes are shaped like rectangular prisms, as shown below. She wants to cover all the sides of each box with special wallpaper. If she has a total of $1120\ \text{ft}^2$ of wallpaper, how many boxes can she cover?



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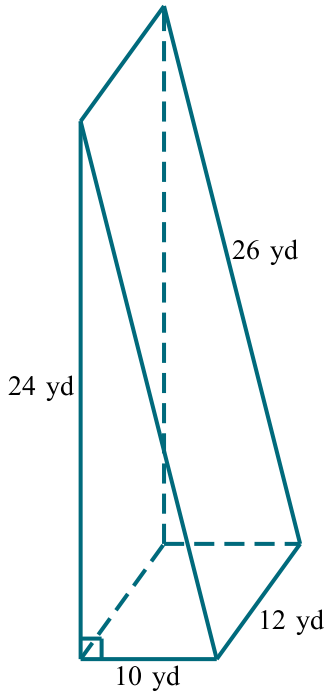
Find the surface area of this triangular prism. Be sure to include the correct unit in your answer.



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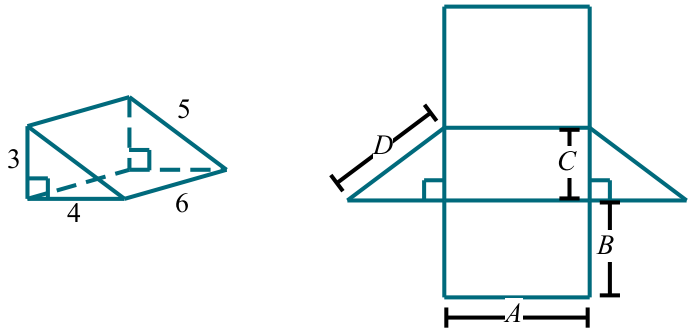
A right triangular prism and its net are shown below.
(All lengths are in centimeters.)



(a) Find the following side lengths for the net.

$A =$ _____cm
$B =$ _____cm
$C =$ _____cm
$D =$ _____cm

(b) Use the net to find the surface area of the prism.

_____cm^2

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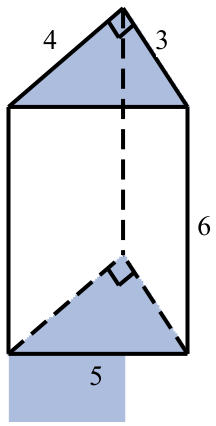
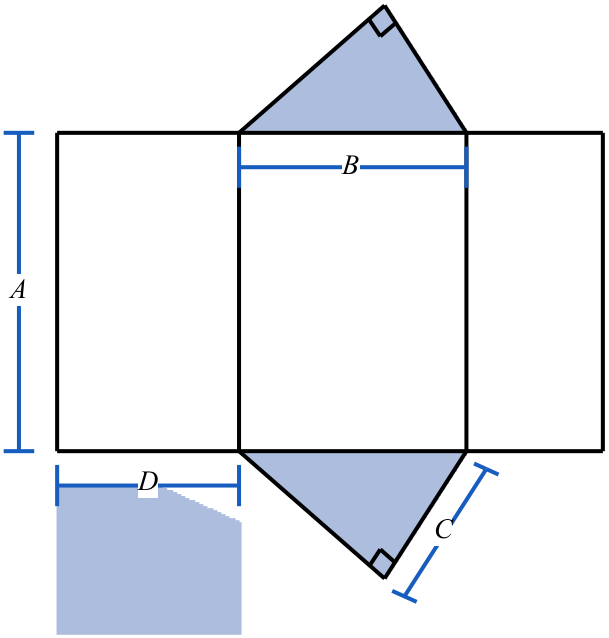
Question 30 of 39

A triangular prism and its net are shown below.
The top and bottom of the prism are shaded.
(All lengths are in centimeters.)





(a) Find the following side lengths for the net.

$A =$ _____ cm
$B =$ _____ cm
$C =$ _____ cm
$D =$ _____ cm

(b) Use the net to find the *lateral* surface area of the prism. Neither the top nor bottom is included.

_____ cm^2

(c) Use the net to find the *total* surface area of the prism.

_____ cm^2

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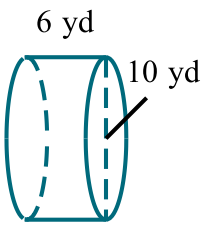
Question 31 of 39

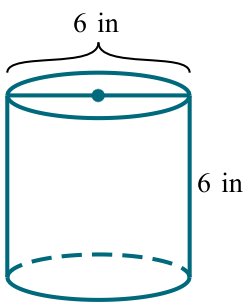
Find the surface area of a cylinder with a height of $6\ \text{yd}$ and a base diameter of $10\ \text{yd}$.

Use the value 3.14 for π, and do not do any rounding.
Be sure to include the correct unit.



Question 32 of 39

A company makes paper labels for paint cans. As shown below, each can is in the shape of a cylinder with a height of $6\ \text{in}$ and a diameter of $6\ \text{in}$. The paper label is wrapped around the can and covers only the side of the can (not the top or bottom). If the company makes 17 labels, how many square inches of paper will be needed? Use 3.14 for π, and do not round your answer.



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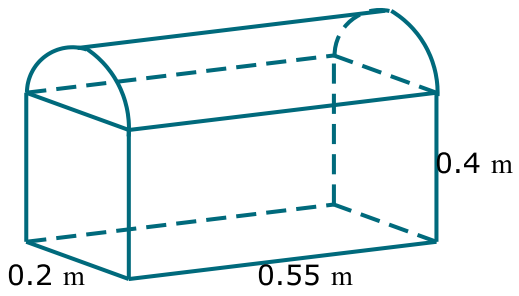
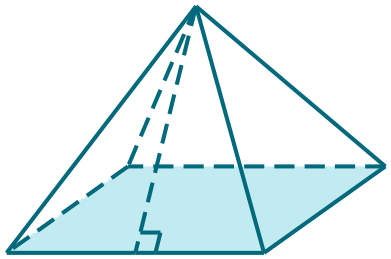
Question 33 of 39

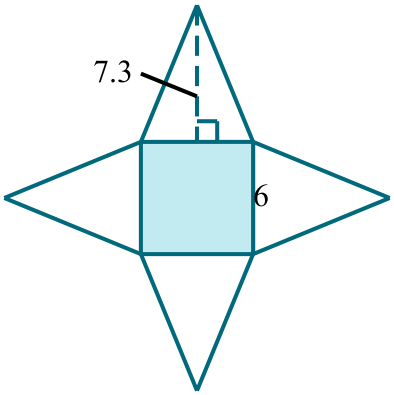
A company manufactures aluminum mailboxes in the shape of a box with a half-cylinder top. The company will make 1384 mailboxes this week. If each mailbox has dimensions as shown in the figure below, how many square meters of aluminum will be needed to make these mailboxes? In your calculations, use the value 3.14 for π, and round up your answer to the next square meter.



Question 34 of 39

Here is a square pyramid and its net.
The lateral faces are congruent triangles. The base (shaded) is a square.
(All lengths are in centimeters.)





(a) Find the area of the base of the pyramid.

_____ cm^2

(b) Find the area of one lateral face of the pyramid.

_____ cm^2

(c) Use the net to find the *lateral* surface area of the pyramid. The base is not included.

_____ cm^2

(d) Use the net to find the *total* surface area of the pyramid.

_____ cm^2

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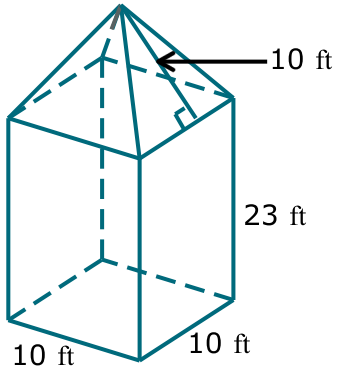
Question 35 of 39

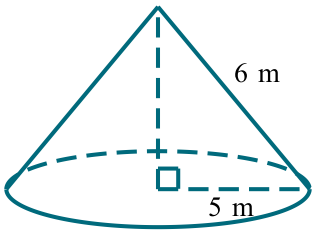
A monument outside city hall has dimensions as shown in the figure below. If one gallon of paint can cover $275\ \text{ft}^2$, how many gallons of paint must be bought in order to paint the monument? Assume that the base of the monument cannot be painted and that paint can be bought only by the gallon.



Question 36 of 39

Find the lateral surface area and surface area of a cone with a slant height of $6\ \text{m}$ and a base radius of $5\ \text{m}$.

Use the value 3.14 for π, and do not do any rounding.
Be sure to include the correct unit in your answers.



(a) lateral surface area:

(b) surface area:

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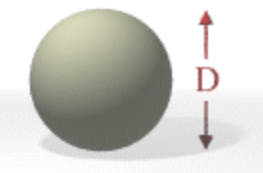
Question 37 of 39

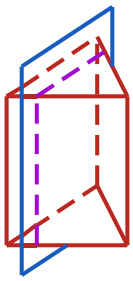
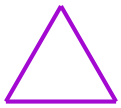
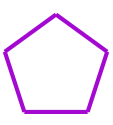
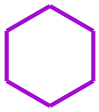
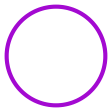
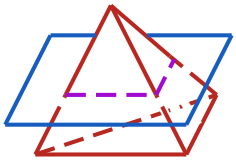
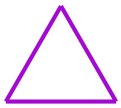
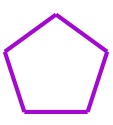
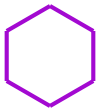
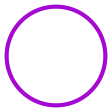
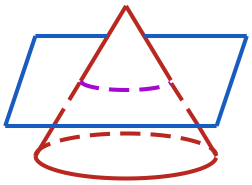
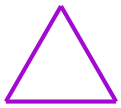
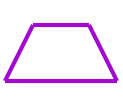
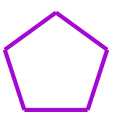
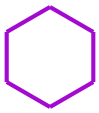
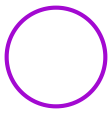
The diameter D of a sphere is $11.6\ \mathrm{m}$. Calculate the sphere's surface area A.

Use the value 3.14 for π, and round your answer to the nearest tenth.



Question 38 of 39

In each figure below, a plane passes through a geometric solid as shown.
For each figure, choose the shape of the cross section formed.

Figure	Cross section					
(a) 					
(b) 					
(c) 					

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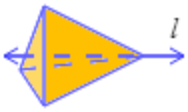
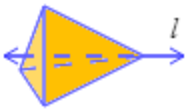
Each region below is being rotated about the line l.
For each region, identify the solid generated by the rotation.

Region	Solid				
(a)					
(b)					
(c)					

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **39**

Question 1 of 39

$200.96\ \text{cm}^2$

Question 2 of 39

$84\ \text{yd}^2$

Question 3 of 39

Area: 12 square units

Perimeter: 14 units

Question 4 of 39

45 square units

Question 5 of 39

$44\ \text{in}^2$

Question 6 of 39

Area: $270\ \text{mm}^2$

Question 7 of 39

Area: $240\ \text{mm}^2$

Question 8 of 39

44 square units

Question 9 of 39

$41.5\ \text{cm}^2$

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Question 10 of 39

$130.2\ \text{yd}^2$

Question 11 of 39

Area $= 18\sqrt{3}\ \text{cm}^2$

Question 12 of 39

$50.24\ \text{m}^2$

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Question 13 of 39

Part 1:
The height of the "parallelogram" is {twice, half, equal to} the radius of the circle.
So, the height of the "parallelogram" is $h =$ _____ cm.

Part 2:
The length of the base of the "parallelogram" is {twice, half, equal to} the circumference of the circle.

The circumference of the circle is _____ cm.
So, the length of the base of the "parallelogram" is $b =$ _____ cm.

Part 3:
Use the parts above to find the area of the "parallelogram".

Area of "parallelogram" = _____ cm^2

Part 4:
The circle was rearranged to make the "parallelogram".
So the area of the circle is equal to the area of the "parallelogram".
Using this fact, choose the equation that gives the area of the circle in terms of r.

- ○ Area $= 9\pi r$
- ○ Area $= \pi r^2$
- ○ Area $= 6\pi r^2$
- ○ Area $= \pi r$
- ○ Area $= 3\pi r^2$

Question 14 of 39

$8039.54\ \text{m}^2$

Question 15 of 39

$160.14\ \text{m}^2$

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Question 16 of 39

139 bags of sand

Question 17 of 39

$14.25\ \text{ft}^2$

Question 18 of 39

$7.74\ \text{m}^2$

Question 19 of 39

$108\sqrt{3} - 36\pi\ \text{yd}^2$

Question 20 of 39

$\frac{275\pi}{36}\ \text{yd}^2$

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Question 21 of 39

(a) Find the ratio of the area of the sector to the area of the circle.

$$\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{\mathbf{1}}{\mathbf{6}}$$

(b) From the figure we see that $m\angle ACB = 60°$.

The measure of $\angle ACB$ is [**equal to**] the measure of $\overset{\frown}{AB}$.

So, $m\overset{\frown}{AB} = \mathbf{60°}$.

This gives the following equation.

$$\frac{m\overset{\frown}{AB}}{360°} = \frac{\mathbf{1}}{\mathbf{6}}$$

(c) Choose the equation that is true.

- (●) $\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{m\overset{\frown}{AB}}{360°}$
- ( ) $\frac{\text{Area of sector}}{\text{Area of circle}} = 6 \cdot \frac{m\overset{\frown}{AB}}{360°}$
- ( ) $\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{360°}{m\overset{\frown}{AB}}$
- ( ) $\frac{\text{Area of sector}}{\text{Area of circle}} = m\overset{\frown}{AB}$

(d) Using πr^2 for the area of the circle, choose the equation that gives a formula for the area of the sector.

- ( ) Area of sector $= \frac{360°}{m\overset{\frown}{AB}} \cdot \pi r^2$
- (●) Area of sector $= \frac{m\overset{\frown}{AB}}{360°} \cdot \pi r^2$
- ( ) Area of sector $= \frac{m\overset{\frown}{AB}}{360°} \cdot 6\pi r^2$
- ( ) Area of sector $= m\overset{\frown}{AB} \cdot \pi r^2$

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Question 22 of 39

Length of $\overset{\frown}{AB}$: $\frac{5}{6}\pi$ m

Area of shaded region: $\frac{5}{4}\pi$ m^2

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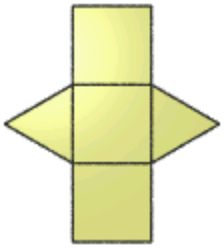
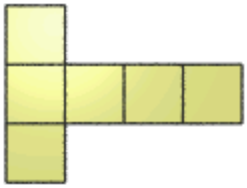
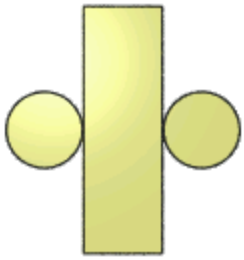
1 a b c None	2  a b c None
3  a b c None	4  a b c None

Question 24 of 39

486 yd^2

Question 25 of 39

(a) Find the following side lengths for the net.

$A = 7$ ft
$B = 10$ ft
$C = 2$ ft
$D = 2$ ft

(b) Use the net to find the surface area of the prism.

208 ft^2

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Question 26 of 39

(a) Find the following side lengths for the net.

$A = 5$ m

$B = 9$ m

$C = 2$ m

$D = 2$ m

(b) Use the net to find the *lateral* surface area of the prism. Neither the top nor bottom is included.

126 m^2

(c) Use the net to find the *total* surface area of the prism.

146 m^2

Question 27 of 39

7 boxes

Question 28 of 39

960 yd^2

Question 29 of 39

(a) Find the following side lengths for the net.

$A = 6$ cm
$B = 4$ cm
$C = 3$ cm
$D = 5$ cm

(b) Use the net to find the surface area of the prism.

84 cm^2

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Question 30 of 39

(a) Find the following side lengths for the net.

$A = 6$ cm

$B = 5$ cm

$C = 3$ cm

$D = 4$ cm

(b) Use the net to find the *lateral* surface area of the prism. Neither the top nor bottom is included.

72 cm^2

(c) Use the net to find the *total* surface area of the prism.

84 cm^2

Question 31 of 39

345.4 yd^2

Question 32 of 39

1921.68 in^2

Question 33 of 39

1266 m^2

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Question 34 of 39

(a) Find the area of the base of the pyramid.

$36\ \text{cm}^2$

(b) Find the area of one lateral face of the pyramid.

$21.9\ \text{cm}^2$

(c) Use the net to find the *lateral* surface area of the pyramid. The base is not included.

$87.6\ \text{cm}^2$

(d) Use the net to find the *total* surface area of the pyramid.

$123.6\ \text{cm}^2$

Question 35 of 39

5 gallons

Question 36 of 39

(a)	lateral surface area: $94.2\ \text{m}^2$
(b)	surface area: $172.7\ \text{m}^2$

Question 37 of 39

$A = 422.5\ \text{m}^2$

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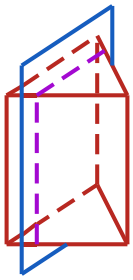
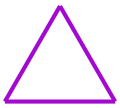
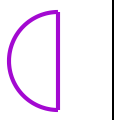
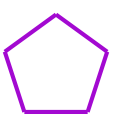
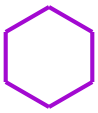
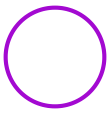
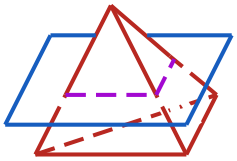
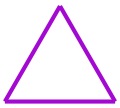
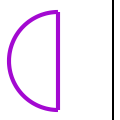
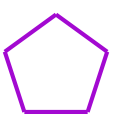
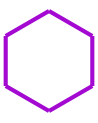
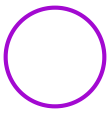
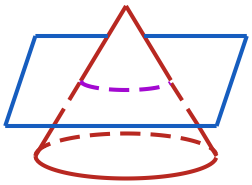
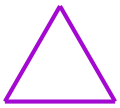
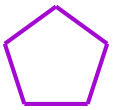
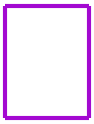
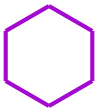
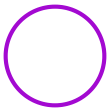
Figure	Cross section					
(a) 						
(b) 						
(c) 						

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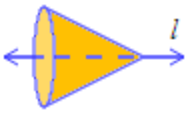
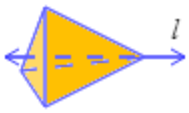
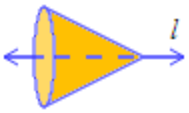
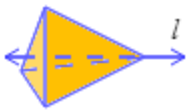
Region	Solid				
(a)					
(b)					
(c)					

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